



Cobalt–boron amorphous alloy prepared in water/cetyl-trimethyl-ammonium bromide/n-hexanol microemulsion as anode for alkaline secondary batteries

D.G. Tong^{a,*}, D. Wang^a, W. Chu^{b,*}, J.H. Sun^a, P. Wu^a

^a College of Materials and Chemistry & Chemical Engineering, Chengdu University of Technology, Chengdu 610059, China

^b Lab 230, College of Chemical Engineering, Sichuan University, Chengdu 610065, China

ARTICLE INFO

Article history:

Received 21 July 2009

Received in revised form 6 November 2009

Accepted 11 November 2009

Available online 3 December 2009

Keywords:

Co–B amorphous alloy

Anode

Alkaline secondary batteries

Microemulsion

ABSTRACT

Amorphous cobalt–boron (Co–B) with uniform nanoparticles was prepared for the first time via reduction of cobalt acetate by potassium borohydride in the water/cetyl-trimethyl-ammonium bromide/n-hexanol microemulsion system. The sample was characterized by X-ray diffraction, transmission electron microscopy, nitrogen adsorption–desorption, X-ray photoelectron spectroscopy, inductively coupled plasma, cyclic voltammetry, differential scanning calorimetry, temperature-programmed desorption, scanning electron microscopy, charge–discharge test and electrochemical impedance spectra. The results demonstrate that electrochemical activity of the as-synthesized Co–B was higher than that of the regular Co–B prepared in aqueous solution. It indicates that the homogeneous distribution and large specific surface area helped the electrochemical hydrogen storage of the as-synthesized Co–B. Furthermore, the as-synthesized Co–B even had 347 mAh g^{−1} after 50 cycles, while the regular Co–B prepared in aqueous solution only had 254 mAh g^{−1} after 30 cycles at a current of 100 mA g^{−1}. The better cycling performance can be ascribed to its smaller interfacial impedance between electrode and electrolyte.

© 2009 Elsevier Ltd. All rights reserved.

1. Introduction

Secondary alkaline batteries play an important role in meeting the growing energy and power demands of consumer devices [1]. Therefore, there is continuous need to search for anode materials to develop secondary alkaline batteries with high energy density. Recently, several research groups have reported that Co–B amorphous alloy is a potential anode for alkaline secondary batteries [2–12]. Many endeavors have been carried out to improve its electrochemical performance. Wang et al. [4] prepared the ultra-fine amorphous Co–B alloy particles with a reversible discharge capacity of about 260 mAh g^{−1} at 300 mA g^{−1}. Meanwhile, Liu et al. [5,6] reported that Co–B alloys prepared by high temperature solid phase process and arc melting have an initial discharge capacity of 480 mAh g^{−1} and can deliver a reversible discharge capacity of about 195 mAh g^{−1} at 25 mA g^{−1}. In addition, Wu et al. proposed that Co–B alloy obtained by ball milling method has a good discharge capacity of 238 mAh g^{−1} at 130 mA g^{−1} [7]. However, for the practice, the electrochemical performance of Co–B alloys is undesirable.

It is well known that the application of nanostructured electrode materials is a desirable approach to improve the electrochemical

performance of batteries [13]. Now, the microemulsion technique has been found to be an effective approach to control synthesis of nanoparticles with uniform distribution [13–16]. The basic concept of the emulsion process is to disperse a solution containing the desired species in an immiscible liquid by adding surfactants and using an emulsifying treatment. Tiny liquid micelles can be well dispersed in the immiscible liquid and each acts as an independent microreactor for the formation of nanoparticles. Meanwhile, the surfactants adsorbed on the formed particles inhibit the aggregation of particles. Li et al. recently successfully synthesized a series of Co–B amorphous alloy catalysts in a cyclohexane/polyethylene glycol/ water microemulsion [17].

In this work, amorphous Co–B was prepared for the first time via reduction of cobalt acetate by potassium borohydride in the microemulsion system of water/cetyl-trimethyl-ammonium bromide/n-hexanol. The electrochemical properties of the as-prepared Co–B as anode for alkaline secondary batteries were examined, and compared with those of the regular Co–B prepared in aqueous solution.

2. Experimental

2.1. Co–B preparation

In the preparation, 20 mL cobalt acetate (0.52 mol L^{−1}) solution was added to a mixture of cetyl-trimethyl-ammonium bromide (32 g) and n-hexanol mixture (48 g) at 298 K, and was well mixed

* Corresponding authors. Tel.: +86 28 8407 8939; fax: +86 28 8407 9074.

E-mail addresses: tongdongge@163.com (D.G. Tong), chuwei65@yahoo.com.cn (W. Chu).

by stirring for 45 min. After homogenization of the mixture, 20 mL of 2 mol L^{-1} potassium borohydride aqueous solution containing 0.22 mol L^{-1} NaOH was added dropwise (2.5 mL min^{-1}) under stirring at ice-bath. When bubbles were no longer released, the black precipitate was washed with distilled water three times, followed by ethanol three times. The as-prepared material was collected and washed with ethanol five times and finally kept in absolute ethanol until use. The as-prepared Co-B sample was designated Co-B-w/o, with w/o representing the water/cetyl-trimethyl-ammonium bromide/n-hexanol microemulsion system.

The direct reduction of cobalt acetate with KBH_4 in pure aqueous solution resulted in the regular Co-B sample [18–21] and was designated Co-B-regular.

2.2. Characterization

X-ray diffraction (XRD) patterns of the samples were acquired on D/rmax-rA X-ray diffractometer with Cu $\text{K}\alpha$ sources (40 kV, 150 mA). The compositions of the samples were analyzed by inductively coupled plasma (ICP, Iris, Advantage). Thermal stability was determined by differential scanning calorimetry (DSC, DSC-131, Setaram) under argon atmosphere, at a heating rate of 10 K min^{-1} . The surface morphologies and particle sizes were observed by means of transmission electron microscopy (TEM, JEOL JSM-200CX). The corresponding mean particle diameters were measured and calculated by counting 200 particles from the enlarged photographs. The specific surface areas were measured by the nitrogen adsorption at 77 K after degassing at 573 K for 3 h. X-ray photoelectron spectroscopy (XPS) was performed on a Perkin-Elmer PHI 5000C ESCA system using Al $\text{K}\alpha$ radiation to determine the surface electronic states of Co-B alloys. All the binding energy (BE) values were calibrated by using $\text{C1s} = 284.6 \text{ eV}$ as a reference.

The cyclic voltammetry was conducted by using a powder microelectrode, which was fabricated according to the procedure described in Ref. [4] by filling the Co-B alloy particles into a micro-cavity at the tip of a Pt microdisk electrode. Meanwhile, a sintered nickel electrode was used as counter electrode, and an Hg/HgO in 30% KOH solution was used as reference electrode.

A porous film electrode embedded with powder was used for the charge–discharge performance measurements of Co-B alloy particles. The Co-B electrode was prepared firstly by mixing 85% Co-B alloy particles, 7% polytetrafluoroethylene (PTFE) and 8% carbon black into paste, then by roll pressing the paste to be film with 0.15 mm thick onto a nickel mesh. The electrochemical measurements were carried out in the laboratory cells using a Co-B electrode (4 cm^2) as anode, a sintered nickel electrode as cathode, and 30% KOH solution as electrolyte.

The temperature-programmed desorption experiments (TPD) for the Co-B electrodes charged and uncharged were carried out in a chemisorption analyzer (Autochem 2, Micromeritics) equipped

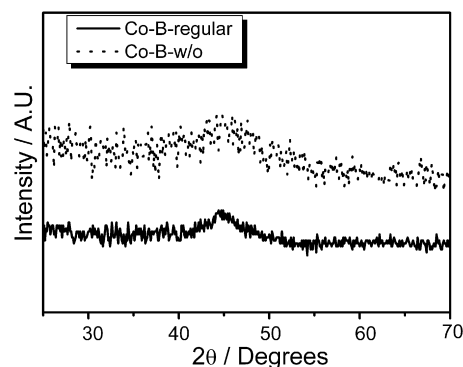


Fig. 1. XRD patterns of Co-B-regular and Co-B-w/o.

with a thermal conductivity detector (TCD). The electrode samples were firstly purged with a flow of purified argon to obtain a stabilized baseline, and then scanned from room temperature to 873 K at a heating rate of 10 K min^{-1} , using argon as a carrier gas. The surface morphologies of the cycled Co-B electrodes were observed by scanning electron microscopy (SEM, JEOL JSM-6500 FE). The electrochemical impedance spectra (EIS) of cells were recorded on the CHI-604A with an alternate current oscillation of 5 mV over the frequency of 0.001 Hz to 100 kHz.

3. Results and discussion

3.1. Structural features

The XRD patterns presented in Fig. 1 show that, similar to the Co-B-regular [18–21], the Co-B sample synthesized in the water/cetyl-trimethyl-ammonium bromide/n-hexanol microemulsion system exhibited a single broad peak around $2\theta = 45.5^\circ$ indicative of amorphous structure [18–21]. The bulk composition of Co-B-w/o is $\text{Co}_{62.9}\text{B}_{37.1}$, which is similar to that of Co-B-regular ($\text{Co}_{63.0}\text{B}_{37.0}$) [18–21].

The TEM images (Fig. 2) show that Co-B-regular displayed irregular, broadly dispersed particles, apparently due to the particle agglomeration, because the reaction between Co^{2+} and BH_4^- is strongly exothermic [17]. In contrast, the Co-B nanoparticles synthesized in the water/cetyl-trimethyl-ammonium bromide/n-hexanol microemulsion system were spherical and uniform. Furthermore, the corresponding mean particle diameters decreased from 37 to 12 nm. The narrow distribution for Co-B-w/o nanoparticles is attributed to the dispersing effect of oil droplets in microemulsion [16,17]. The specific surface area of Co-B-w/o ($163.23 \text{ m}^2 \text{ g}^{-1}$) was much larger than that of Co-B-regular ($65.00 \text{ m}^2 \text{ g}^{-1}$), which is due to the formation of the homogeneous fine particles in Co-B-w/o [16,17].

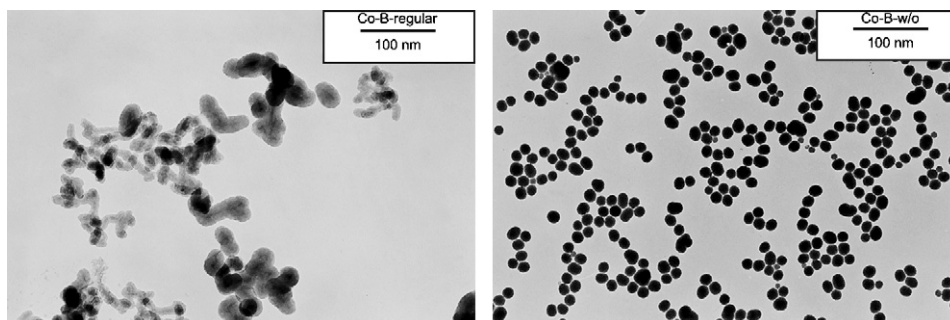


Fig. 2. TEM images of Co-B-regular and Co-B-w/o.

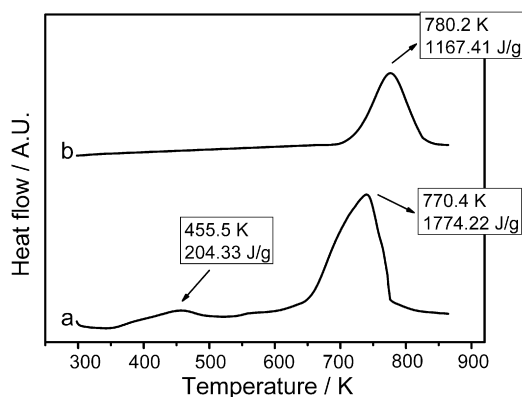


Fig. 3. DSC curves of (a) Co-B-regular and (b) Co-B-w/o.

The DSC analysis (Fig. 3a) indicates that the crystallization of the Co-B-regular involved two steps [17,22]: rearrangement of amorphous alloy structure and crystallization with exothermic peaks at around 455.5 K (204.33 J g^{-1}) and 770.4 K (1774.22 J g^{-1}). However, Co-B-w/o displayed only one exothermic peak at 780.2 K, higher than that of Co-B-regular by 10 K. Furthermore, it released less heat (1167.41 J g^{-1}). Obviously, Co-B-w/o exhibited much higher thermal stability against crystallization than Co-B-regular. The enhanced thermal stability is attributed to the uniform particle size of Co-B-w/o, which may inhibit the particle migration and agglomeration [17,22].

The XPS spectra (Fig. 4) revealed that for all Co species in either Co-B-regular or Co-B-w/o, the core level of Co 2p_{3/2} was at 778.6 eV, indicating that all Co atoms were present in metallic state [17–23]. The B species in Co-B-regular were present in the elemental state (188.4 eV), while the B species in Co-B-w/o were present in both the elemental state (188.4 eV) and oxidized state (192.1 eV). One possible reason was that the particles of Co-B-w/o were too fine to prevent from being exposed to oxygen during the pretreatment of samples before XPS analysis [16]. The fact that the oxidized B species could not be detected by XRD suggested that the oxidized B species in Co-B-w/o is amorphous [18]. Contrary to the B species, no significant oxidation of the metallic Co occurred through the preparation process, indicating that the existence of the B species may effectively protect the metallic Co from oxidation.

3.2. Cyclic voltammetry and electrochemical hydrogen storage

Fig. 5 shows the cyclic voltammograms of Co-B-regular and Co-B-w/o electrodes at a scanning rate of 1 mV s^{-1} . Both samples exhibited a couple of a cathodic and an anodic peak, indicating that the electrochemical reaction occurring on the Co-B electrodes is reversible. Similar to previous reports [4,7], Co-B-regular had a

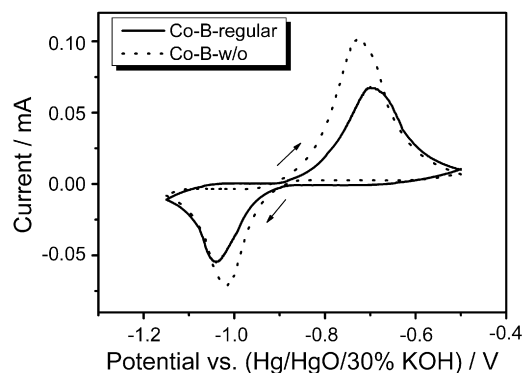


Fig. 5. Cyclic voltammograms of Co-B-regular and Co-B-w/o electrodes at a scanning rate of 1 mV s^{-1} .

cathodic peak at -1.03 V and an anodic peak at -0.71 V . Co-B-w/o demonstrated lower anodic peak and higher cathodic peak, respectively (Fig. 5). The decrease of hysteresis between the anodic and cathodic peaks indicates that the charge–discharge reversibility of Co-B alloy particles was improved [8,24–26]. Since the cathodic peaks are very similar to the CV features for electrochemical reduction of H_2O in the potential position and peak shape, it is most likely that the cathodic peaks in the CV curves arise from the hydrogen adsorption on the Co-B electrodes [1–4]. Since the equilibrium potentials of Co and B elements are at -0.83 and -1.81 V (versus Hg/HgO) [1], respectively, it is difficult to assign the anodic current peaks to the oxidation of elemental Co or B. The close resemblance of these anodic peaks to the CV features of electrochemical oxidation of hydrogen, as observed previously on various hydrogen storage electrodes [1–4] strongly suggests that the anodic peaks in Fig. 5 arise mainly from the electrochemical oxidation of adsorbed hydrogen. Of course, the anodic oxidation of Co-B could not be completely excluded. The fact that the reduction peaks are smaller than the oxidation peaks may be ascribed to the simultaneous oxidation of some Co-B to Co(OH)_2 . The larger area of the cathodic peak of Co-B-w/o than that of Co-B-regular indicates that the electrochemical hydrogen storage capacity was improved for Co-B-w/o [1,24]. It was also confirmed by the H_2 -TPD analysis (Fig. 6) for the Co-B-regular and Co-B-w/o electrodes taken from the fully charged cells.

As it can be seen in Fig. 6, the TPD profiles of the uncharged electrodes appeared as flat baselines, showing no sign of hydrogen desorption from the electrodes. However, for the fully charged electrodes, huge desorption peaks appeared. It further confirms the electrochemical hydrogen storage behavior of the Co-B particles [1].

The H_2 -TPD curve of Co-B-regular (Fig. 6) showed seven distinct desorption peaks at 517, 599, 625, 676, 754, 824 and 844 K. For

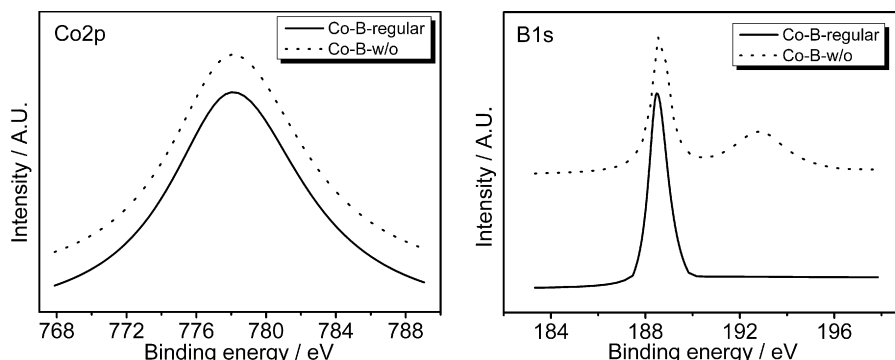


Fig. 4. Co2p_{3/2} and B1s XPS spectra of Co-B-regular and Co-B-w/o.

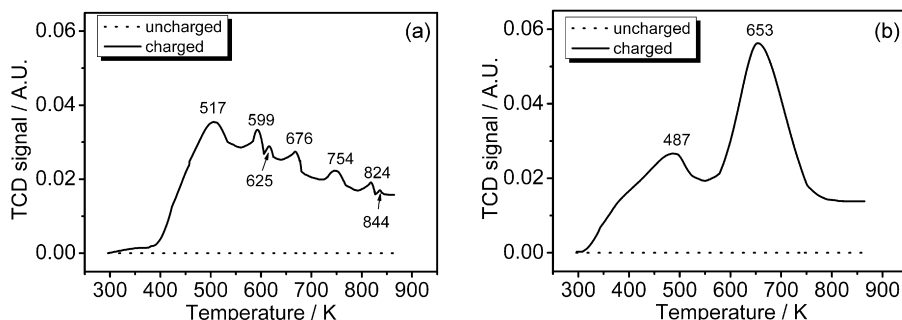


Fig. 6. TPD profiles of (a) Co-B-regular and (b) Co-B-w/o electrodes taken from the fully charged cells and uncharged cells.

Co-B-w/o, only a single intense peak at 653 K and a small shoulder peak at 487 K were observed. It suggests that Co-B-w/o contained uniform electrochemical active sites [1,24].

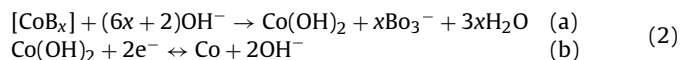
By calculating the total area of the TPD peaks in Fig. 6, the amount of hydrogen released from the Co-B-regular electrode is 83.45 mL g^{-1} , corresponding to a discharge capacity of 213 mAh g^{-1} , which is considerably lower than the realized discharge capacity (296 mAh g^{-1}) of the Co-B-regular [1,4]. This capacity loss is probably arisen from the release or oxidation of hydrogen atoms when the electrode was taken from the cell and exposed in air during experiment [1,4].

The calculated total area of the TPD peaks in Fig. 6b for Co-B-w/o is 95.33 mL g^{-1} , which is larger than that of Co-B-regular. It indicates that electrochemical hydrogen adsorption was stronger on Co-B-w/o than on Co-B-regular [16], which is the same as the result obtained in CV analysis.

3.3. Charge–discharge curves and cycling performance

To examine the reversible electrochemical capacity of the Co-B alloys, we constructed the experimental alkaline secondary cells using the Co-B negative electrodes and sintered nickel positive electrodes in 30% KOH solution, and cycled the cells at conventional charge–discharge conditions. Fig. 7 shows the charge–discharge curves of the alkaline secondary batteries at a constant current of 100 mA g^{-1} . For both samples, the charging and discharging voltage plateaus appeared at ~ 1.35 and ~ 1.15 V, respectively, which are characteristic of the reversible electrochemical storage and oxidation of hydrogen [4].

Co-B-regular showed an initial discharge capacity of 296 mAh g^{-1} , which is similar to those reported for Co-B alloy [4,7]. In the recent works [4–12], people reported that Co-B might react as a negative electrode in the secondary alkaline batteries in two formats:



It has been well recognized that cobalt can form a solid solution with hydrogen with the atomic ratio of hydrogen to cobalt up to 1. If the Co-B alloy is hydrogenated to form Co_2BH_2 , the maximal discharge capacity in the former reaction is 414 mAh g^{-1} , while the capacity for $\text{Co/Co}(\text{OH})_2$ in the latter reaction is not more than 80 mAh g^{-1} . From CV tests, both process (1) and process (2) are involved in this experiment. Therefore, it is important to clarify which reaction takes place and contributes to the high discharge capacity. Based on previous works [1,4], Co and B do not account for the high discharge capacity. Furthermore, $\text{Co}(\text{OH})_2$ was produced only at the more positive potential in late stage of discharge, it can contribute only a small part to the discharge capacity of the Co-B electrodes [1]. Thus, the discharge capacities of Co-B cannot be simply accounted for the reaction of $\text{Co/Co}(\text{OH})_2$ in this work.

As a part of Co-B alloy in the electrodes will be oxidized to $\text{Co}(\text{OH})_2$ in the alkaline solution during the charge–discharge process, which cannot form hydride and would certainly lead to a reduction of the observed capacity from its theoretical value. Thus, the observation of 296 mAh g^{-1} reversible capacity for the Co-B alloy particles is reasonable.

From the capacity analysis above, some Co-B may contribute to the high discharge capacity via process (2). However, the reaction of Co-B electrodes in this work should be mainly attributed to process (1), that is, the hydrogenation–dehydrogenation takes place in the electrode. We have to admit that at present time it is difficult to make clear of the relation between process (1) and process (2) during charge–discharge process. We hope further investigations help to solve the problems and accelerate its practical application in commercial batteries.

The Co-B-w/o showed an activation process from the initial reversible capacity of 325 mAh g^{-1} to its highest capacity of 357 mAh g^{-1} at the second cycle. The fact of higher discharge capacity of Co-B-w/o than that of Co-B-regular further confirms

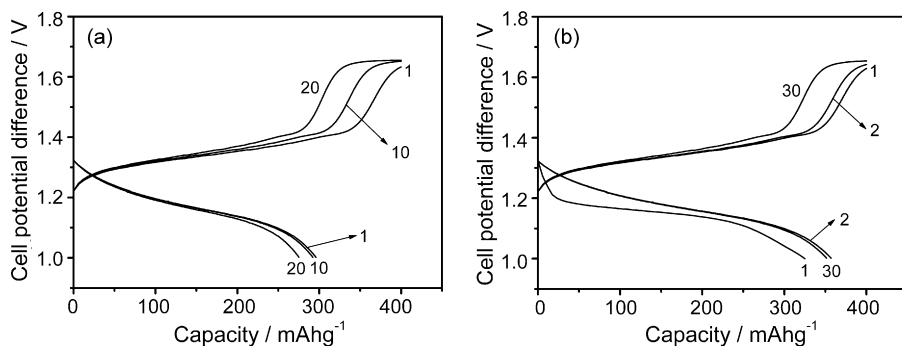


Fig. 7. Charge–discharge curves of Co-B-regular and Co-B-w/o at a constant current of 100 mA g^{-1} in different cycles.

that the electrochemical activity of Co-B-w/o was higher than that of Co-B-regular. The larger potential drop of Co-B-w/o at the initial cycle might be attributed to its activation [8,24,25] resulted from the formation of the small amount of oxidized B species as shown by XPS. At the initial charge–discharge stage, the formed oxidized B species hindered surface activation of Co-B alloy electrodes. Therefore, it results in a large charge-transfer resistance in the first cycle (Section 3.4), which causes the decrease of electrode potential plateau and leads to the decrease of the discharge capacity (Fig. 7). With the reaction, the oxidized B species on the surface dissolved to the alkaline solution, which activate the electrode surface, thus, the observation of its highest reversible capacity of 357 mAhg^{-1} at the second cycle is reasonable.

Fig. 8 shows the cycling performance of Co-B-regular and Co-B-w/o at a constant current of 100 mA g^{-1} . After 1 cycle, the discharge capacity of Co-B-w/o increased to 357 mAhg^{-1} . It indicates that the Co-B-w/o had activation in the initial cycle [8,24,25]. Then, Co-B-w/o exhibited excellent cycling performance. It still maintained 346 mAhg^{-1} after 50 cycles, while Co-B-regular only kept 246 mAhg^{-1} after 30 cycles. The excellent cycling performance for

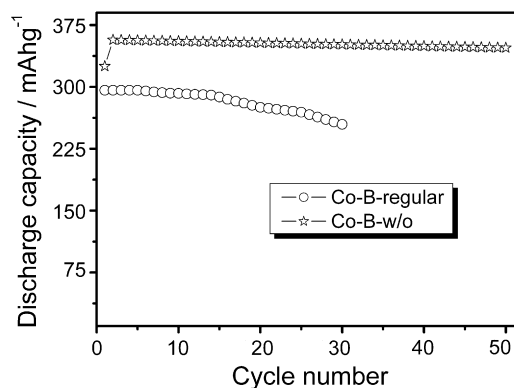


Fig. 8. Cycling performance of Co-B-regular and Co-B-w/o at a constant current of 100 mA g^{-1} .

Co-B-w/o is attributed to its smaller interfacial impedance between electrode and electrolyte as shown by EIS analysis (Section 3.4).

SEM micrographs of cycled electrodes after different cycles (Fig. 9) revealed that partial crystallization occurred for all elec-

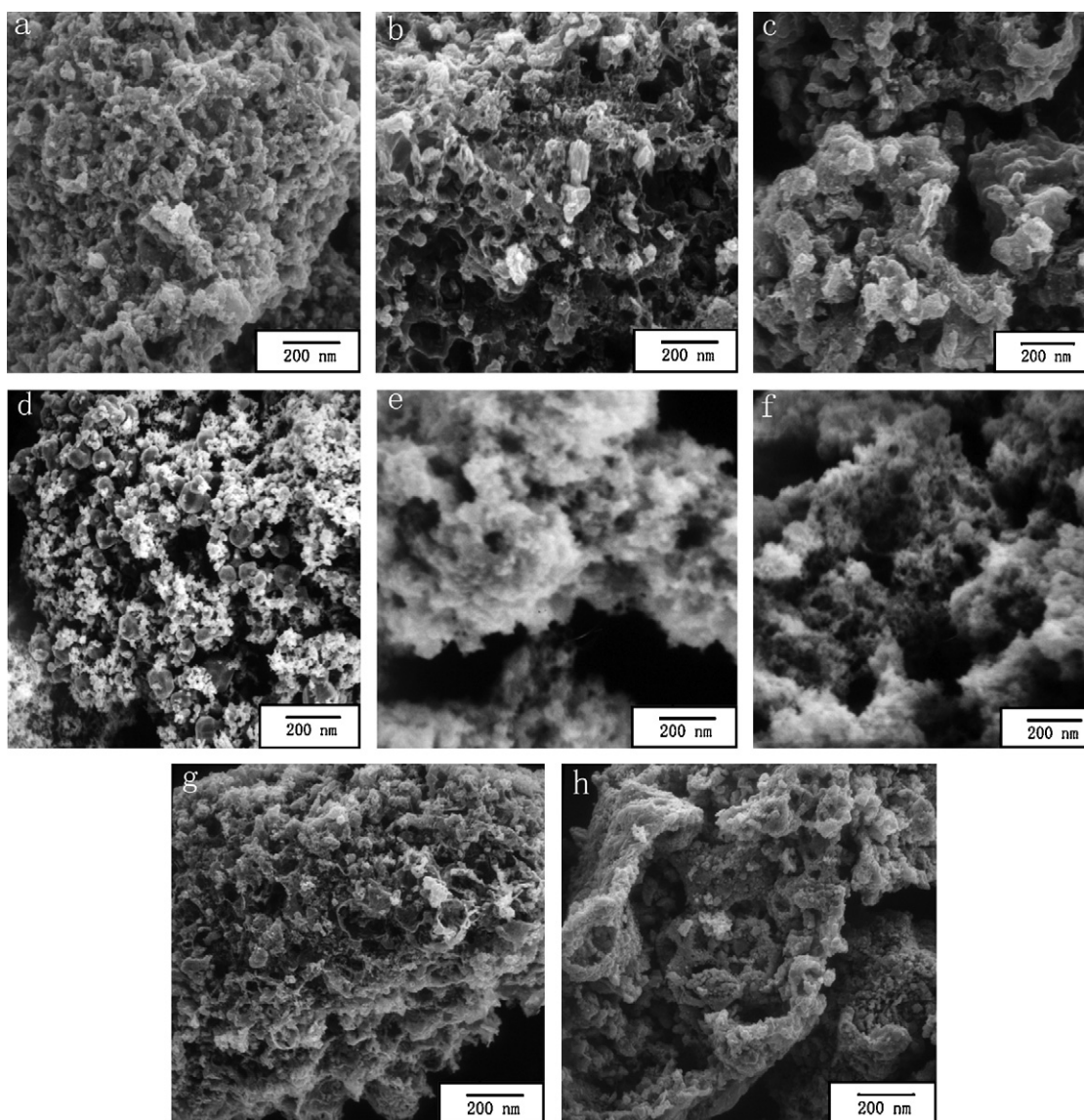


Fig. 9. SEM images of Co-B-regular electrodes after (a) 1 cycle, (b) 10 cycles, (c) 20 cycles, (d) 30 cycles, and Co-B-w/o electrodes after (e) 1 cycle, (f) 10 cycles, (g) 30 cycles, and (h) 50 cycles.

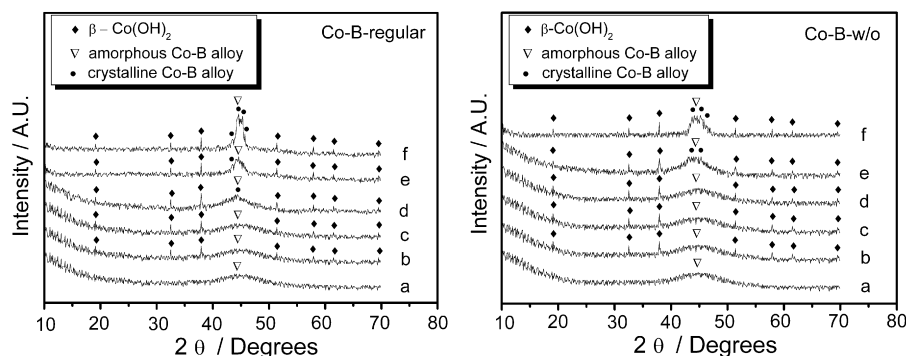


Fig. 10. XRD patterns of Co-B-regular electrodes at (a) fully charged state of the initial cycle, (b) fully discharged state of the initial cycle, (c) fully discharged state after 2 cycle, (d) fully discharged state after 10 cycles, (e) fully discharged state after 20 cycles, (f) fully discharged state after 30 cycles, and Co-B-w/o electrodes at (a) fully charged state of the initial cycle, (b) fully discharged state of the initial cycle, (c) fully discharged state after 2 cycle, (d) fully discharged state after 10 cycles, (e) fully discharged state after 30 cycles, (f) fully discharged state after 50 cycles.

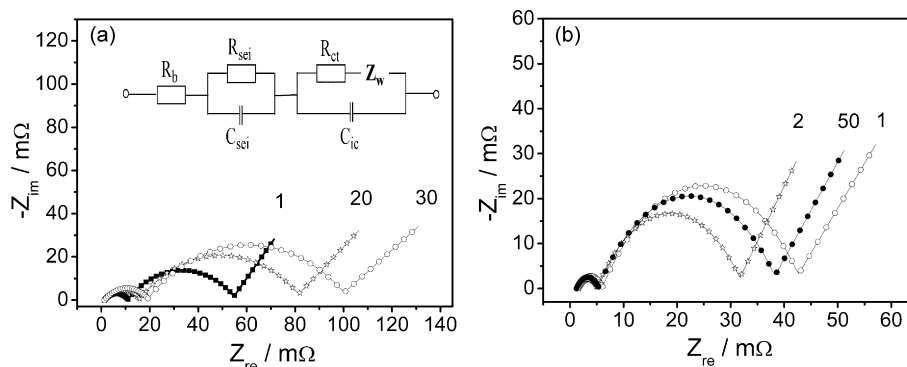


Fig. 11. EIS spectra of (a) Co-B-regular and (b) Co-B-w/o in different cycles.

trodes, which is responsible for the decrease of discharge capacity. However, the crystallization of Co-B-w/o is not significant. It is attributed to its high thermal stability and homogeneous distribution. The small holes on the surfaces (Fig. 9) are possible due to the structure shrinkage of Co-B amorphous alloy caused by the occurrence of partial crystallization during the cycling. As a result, the specific surface areas and the active site of electrochemical hydrogen storage may have decreased due to the agglomeration of particles [17,22].

In order to confirm the occurrence of partial crystallization of Co-B amorphous alloy during cycling, we also used XRD to characterize the changes in the structure of Co-B electrodes after different cycles. When the Co-B electrodes were charged firstly to fully charged state at a current of 100 mA g^{-1} , the XRD patterns remained almost unchanged, showing no structural change occurring during charging process. However, $\beta\text{-Co(OH)}_2$ (JCPDS No. 30-443) appeared after the electrodes were discharged. At following charge–discharge process, these XRD peaks of $\beta\text{-Co(OH)}_2$ were still present (Fig. 10). Based on the CV tests and XRD analysis, Co(OH)_2 may arise from the oxidation of Co-B electrodes in alkaline solution during charge–discharge cycling.

Meanwhile, the crystallization of Co-B amorphous alloy was observed in the XRD patterns of the cycled electrodes. From Fig. 10, it can be seen that Co-B-w/o effectively inhibited the crystallization during cycling [17,22,26], which further confirms the results obtained from SEM analysis (Fig. 9).

Based on the above experimental results, it can be concluded that the grain growth in the electrodes after several cycles observed by SEM analysis (Fig. 9) is attributed to the partial crystallization of Co-B amorphous alloy during cycling. Therefore, the deactivation of Co-B amorphous alloy during cycling can be attributed to the occurrence of crystallization.

3.4. EIS spectra analysis

Fig. 11 demonstrates the EIS spectra of Co-B-regular and Co-B-w/o after different cycles, respectively. For all the spectra, high-frequency semicircles, intermediate-frequency semicircles, and low frequency plots are observed [24,25]. In general, the high-frequency intercept on the real axis indicates the bulk resistance (R_b) of the cell, which are primary the resistance of the electrolyte [24]. According to the equivalent circuit of the electrode impedance for secondary alkaline batteries [24,25], the high-frequency semicircle is associated with the resistance (R_{sei}) and capacitance (C_{sei}) between the alloy particles and the current collector. The intermediate-frequency semicircle reflects the charge-transfer resistance (R_{ct}) for electrochemical reaction at the surface and interfacial capacitance between the electrolyte and electrodes (C_{ic}), while the sloping line at low frequencies reflects diffusion impedance in the solid-state electrodes (Z_w). Simplified equivalent circuit model is constructed to analyze the impedance spectra in the inset of Fig. 11a.

The parameters of the equivalent circuit (R_b , R_{sei} , and R_{ct}) for Co-B-regular and Co-B-w/o are shown in Table 1. Except for the resistance of Co-B-w/o in the first cycle, all the values of resistance show an increasing trend with increase in the number of cycles. The higher resistance of Co-B-w/o in the first cycle is due to its activation [1,8,24,25], which is confirmed by the charge–discharge tests (Figs. 7 and 8). With subsequent cycling, the dissolution of oxidized B species in the alkaline solution during the charge–discharge cycles causes fresh interface between Co-B-w/o electrodes and electrolyte and decreases the resistance (Fig. 11).

From Fig. 11 and Table 1, it is obvious that the pronounced difference appeared at the intermediate-frequency semicircle after different cycles, indicating that the charge-transfer resistance is the

Table 1

Impedance parameters of Co–B-regular and Co–B-w/o amorphous alloy electrodes in the different cycles.

Parameters	Co–B-regular			Co–B-w/o		
	1st cycle	20th cycle	30th cycle	1st cycle	2nd cycle	50th cycle
$R_p/m\Omega$	1.37	1.62	1.81	1.71	1.23	1.56
$R_{sei}/m\Omega$	9.21	13.76	17.93	4.58	3.10	4.23
$R_{ct}/m\Omega$	38.64	61.45	77.69	37.23	25.72	33.34

major contributor to the total electrode resistance. The increase in the charge-transfer resistance with cycling for Co–B electrodes can be attributed to the occurrence of crystallization (Figs. 9 and 10) and the formation of $\text{Co}(\text{OH})_2$ layer during charge–discharge cycling on the Co–B electrodes (Fig. 10). The occurrence of crystallization and the formation of $\text{Co}(\text{OH})_2$ declines the electrochemically activity of the Co–B alloy and inhibits the electrochemical hydrogen reaction on the alloy surface and, consequently, leads to the increase in the charge-transfer resistance and the decrease in the discharge capacity (Figs. 7 and 8). The lower R_{ct} for Co–B-w/o during cycling are possibly attributed to its homogeneous distribution of electrochemical active site for hydrogen storage as confirmed by H_2 -TPD (Fig. 6), and higher thermal stability that inhibited the crystallization during cycling (Figs. 9 and 10).

4. Conclusions

Amorphous Co–B with uniform distribution was successfully prepared via reduction of cobalt acetate by potassium borohydride in the water/cetyl-trimethyl-ammonium bromide/n-hexanol microemulsion system. The electrochemical performance of Co–B-w/o as anode for alkaline secondary batteries was better than that of regular Co–B amorphous alloy prepared in the aqueous solution. It indicates that the homogeneous distribution and large specific surface area helped the electrochemical hydrogen storage of Co–B-w/o particles. Meanwhile, Co–B-w/o even had 347 mAh g^{-1} after 50 cycles, while Co–B-regular only had 254 mAh g^{-1} after 30 cycles at a current of 100 mAh g^{-1} . The better electrochemical performance of Co–B-w/o can be ascribed to its smaller interfacial impedance between electrode and electrolyte.

Acknowledgements

This work was supported by Research Fund of Chengdu University of Technology (grant nos. 2007-YG2 and 2008-071) and Natural

Science Foundation of Sichuan Province (grant no. 07ZA004). The authors would like to thank Engineer K.Y. Zhang for his assistance with the electrochemical performance test and Dr. F. Peng for the TPD analysis in this work.

References

- [1] F. Wu, Novel System of Secondary Batteries, Science Press, Beijing, 2007.
- [2] M. Motiv, A. Popov, I. Dragieva, Colloids Surf. A: Histochem. Eng. Aspects 149 (1999) 413.
- [3] M. Motiv, A. Popov, I. Dragieva, J. Appl. Electrochem. 29 (1999) 59.
- [4] Y.D. Wang, X.P. Ai, H.X. Yang, Chem. Mater. 16 (2004) 5194.
- [5] Y. Liu, Y.J. Wang, L.L. Xiao, D.W. Song, L.F. Jiao, H.T. Yuan, Electrochem. Commun. 9 (2007) 925.
- [6] Y. Liu, Y.J. Wang, L.L. Xiao, D.W. Song, Y.P. Wang, L.F. Jiao, H.T. Yuan, Electrochim. Acta 53 (2008) 2265.
- [7] C. Wu, Y. Bai, F. Wu, L.W. Dong, X. Wang, L.X. Yang, C.Z. Zhang, Electrochim. Acta 53 (2008) 4715.
- [8] D.G. Tong, Y.Y. Luo, W. Chu, Mater. Chem. Phys. 112 (2008) 907.
- [9] C. Wu, Y. Bai, F. Wu, X. Wang, F. Wu, C.Z. Zhang, Solid State Ionics 179 (2008) 924.
- [10] D.W. Song, Y.J. Wang, Y.P. Wang, L.F. Jiao, H.T. Yuan, Electrochem. Commun. 10 (2008) 1486.
- [11] D.G. Tong, W. Chu, X.L. Zeng, W. Tian, D. Wang, Mater. Lett. 63 (2009) 1555.
- [12] D.S. Lu, W.S. Li, X. Jiang, C.L. Tan, R.H. Zeng, J. Alloys Compd. (2009), doi:10.1016/j.jallcom.2009.06.060.
- [13] T. Doi, Y. Iriyama, T. Abe, Z. Ogumi, Chem. Mater. 17 (2005) 1580.
- [14] S. Eriksson, U. Nylen, S. Rojas, M. Boutonnet, Appl. Catal. A 265 (2004) 207.
- [15] D.G. Tong, Q.Y. Lai, N.N. Wei, A.D. Tang, L.X. Tang, K.L. Huang, X.Y. Ji, Mater. Chem. Phys. 94 (2005) 423.
- [16] S.J. Chiang, B.J. Liaw, Y.Z. Chen, Appl. Catal. A 319 (2007) 144.
- [17] H. Li, J. Liu, S.H. Xie, M.H. Qiao, W.L. Dai, H.X. Li, J. Catal. 259 (2008) 104.
- [18] D.G. Tong, W. Chu, Y.Y. Luo, H. Chen, X.Y. Ji, J. Mol. Catal. A: Chem. 269 (2007) 149.
- [19] D.G. Tong, X. Han, W. Chu, H. Chen, X.Y. Ji, Mater. Res. Bull. 43 (2008) 1327.
- [20] D.G. Tong, X. Han, W. Chu, H. Chen, X.Y. Ji, Mater. Lett. 61 (2007) 4679.
- [21] D.G. Tong, J.Y. Hu, W. Chu, T. Zhang, X.Y. Ji, Mater. Lett. 62 (2008) 2746.
- [22] H.X. Li, W.J. Wang, H. Li, J.F. Deng, J. Catal. 194 (2000) 211.
- [23] D.G. Tong, W. Chu, Y.Y. Luo, X.Y. Ji, Y. He, J. Mol. Catal. A: Chem. 265 (2007) 195.
- [24] Q.X. Cha, Kinetics of Electrode, Science Press, Beijing, 2002.
- [25] Y.G. Tang, W.L. Li, Ni-MH Batteries, Chemical Industry Press, Beijing, 2007.
- [26] G.H. Tu, W.R. Chen, Mater. Sci. Eng. 97 (1988) 329.